

Seokhwan Yang

M.S. Student

Graduate School of Metaverse, KAIST

ysshwan147@kaist.ac.kr <https://ysshwan147.github.io/> Google Scholar

Education

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Metaverse

Sep. 2024 – Present

Advisor: Prof. Woontack Woo

Chung-Ang University

B.S. in Computer Science and Engineering, *Cum Laude*

Mar. 2019 – Aug. 2024

Research Experience

M.S. Student Researcher, UVR Lab, KAIST, Republic of Korea

Sep. 2024 – Present

Conducting research on VR/AR avatars and real-time facial expression control.

Developing photorealistic Gaussian-based avatar rendering and control pipelines.

Undergraduate Research Intern, UVR Lab, KAIST, Republic of Korea

Jan. 2024 – Feb. 2024

Explored methods to map HMD-occluded facial expressions to a parametric face model.

Undergraduate Research Intern, VEA Lab, Chung-Ang University, Republic of Korea

Jul. 2023 – Dec. 2023

Performed a comparative study of IK methods for upper-body avatar pose estimation in VR.

Undergraduate Research Intern, VR Lab, Chung-Ang University, Republic of Korea

Jul. 2022 – Aug. 2022

Extended a VR ski simulator to a snowboarding scenario using plantar pressure sensors.

Publications

International

OFERA: Blendshape-driven 3D Gaussian Control for Occluded Facial Expression to Realistic Avatars in VR

Seokhwan Yang, Boram Yoon, Seoyoung Kang, Hail Song, Woontack Woo

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026.

(Presented at IEEE VR 2026)

Streamlined Facial Data Collection based on Utterance and Emotional Data for Human-to-Avatar Reconstruction

Seoyoung Kang, Seokhwan Yang, Hail Song, Boram Yoon, Jinwook Kim, Kangsoo Kim, Woontack Woo

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026.

(Presented at IEEE VR 2026)

VRGaussianAvatar: Integrating 3D Gaussian Avatars into VR

Hail Song, Boram Yoon, Seokhwan Yang, Seoyoung Kang, Woontack Woo

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026.

(Presented at IEEE VR 2026)

HMD-only Controllable 3D Gaussian Avatars in VR: Face and Full-body Demonstration
Seokhwan Yang, Hail Song, Woontack Woo
IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW),
Demo, 2026.

Top 8 Research Demo Nomination

Fast Texture Transfer for XR Avatars via Barycentric UV Conversion

Hail Song, Seokhwan Yang, Woontack Woo

IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct),
Poster, 2025.

Human-to-Avatar Face Representation of Extended Reality (XR) Glasses Wearers for Communication

Seoyoung Kang, Seokhwan Yang, Woontack Woo

IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), XR-
Stand Workshop, 2025.

Domestic

VR Headset User's Blendshape-driven Facial Expression Tracking for Parametric Model and Photo-realistic Avatar Reconstruction

Seokhwan Yang, Woontack Woo

Journal of KIISE, 2026.

(Invited journal extension of KSC 2024 Best Paper)

Speech-to-3D: Personalized 3D Scene Rendering based on User Speech Recognition

Suji Kang, Seokhwan Yang, Seok-Young Kim, Woontack Woo

Korea Computer Congress (KCC), 2025.

VR Headset User's Facial Expression Tracking and Mesh Model Representation for Photo-realistic Head Avatar Generation

Seokhwan Yang, Hyungwoo Jin, Woontack Woo

Korea Software Congress (KSC), 2024.

Best Paper Award

Comparison Study on IK Methods for VR Upper-body Avatar Generation

Seokhwan Yang, Kyoungju Park

HCI Korea, 2024.

Academic Activities

Reviewer

ACM SIGGRAPH 2026

Demo Exhibitor

IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2025

National R&D Project Booth (Open XR platform)

Projects

Professional AI Talent Development Program for Multimodal AI Agents

Graduate Researcher

Supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP)

Grant by the Ministry of Science and ICT(MSIT), Republic of Korea

Sep. 2025 – Present

Development of Open XR platform for high immersive collaboration

Graduate Researcher

Supported by the National Research Council of Science & Technology (NST)

Grant by the Ministry of Science and ICT(MSIT), Republic of Korea

Sep. 2024 – Present